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Video

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6 VIDEO

VIDEO STANDARDS 272

Video Standards

6 VIDEO

VIDEO STANDARDS 273

DV camcorders or VTRs connected to computers over FireWire are used for reasonable quality digital video capture.

Cheap video cameras are often built into mobile phones and laptop computers or used as Webcams. They usually use MPEG-4 and USB 2.0.

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VIDEO STANDARDS 274

Digital video standards inherit features from analogue broadcast TV.

Each frame is divided into two fields (odd and even lines), transmitted one after the other and interlaced for display. Interlaced frames may display combing when displayed progressively or exported as still images.

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VIDEO STANDARDS 275



Interlaced fields

Interlaced video is a technique for doubling the perceived frame rate of a video display without consuming extra bandwidth.

The interlaced signal contains two fields of a video frame captured at two different times.



A screenshot from HandBrake, demonstrating the difference between deinterlaced and interlaced images, from "Deinterlacing Guide". HandBrake.

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VIDEO STANDARDS 276



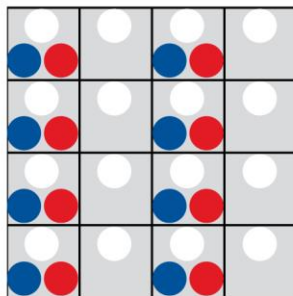
Separated fields and combined frame (right) showing combing

PAL: a frame has 625 lines, of which 576 are picture, displayed at 50 fields (25 frames) per second (625/50).

NTSC: a frame has 525 lines, of which 480 are picture, displayed at 59.94 fields (29.97 frames) per second (525/59.94, often treated as 525/60).

CCIR 601 (Rec. ITU-R BT.601) defines standard definition digital video sampling, with 720 luminance samples and 2×360 colour difference samples per line. ($Y'C_B C_R$ with 4:2:2 chrominance sub-sampling.)

PAL frames are 720×576 and NTSC are 720×480 . The pixels are not square.



4:2:2 chrominance sub-sampling

DV applies 4:1:1 chrominance sub-sampling and compresses to a constant data rate of 25 Mbits per second, a compression ratio of 5:1.

MPEG defines a series of standards. MPEG-2 is used on DVDs; MPEG-4 supports a range of multimedia data at bit rates from 10 kbps to 300 Mbps or greater.

MPEG-4 is a multi-part standard. Part 2 defines a video codec; Part 10 (H.264/AVC) is an improved version.

MPEG standards all define a set of profiles (features) and levels (parameters). The Baseline, Extended and Main profiles of H.264/AVC are all used in multimedia.

MPEG-4 defines a file format. 3GP is a simpler version, used in mobile phones.

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VIDEO STANDARDS 285

HD video uses higher resolutions and may be progressively scanned. Frames with widths of 720 and 1080 pixels and an aspect ratio of 16:9 are used.

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VIDEO COMPRESSION 286

Video Compression

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VIDEO

VIDEO COMPRESSION 287

Spatial (intra-frame) compression and temporal (inter-frame) compression are used together in most contemporary video codecs.

A codec is a device or computer program for encoding or decoding a digital data stream or signal. Codec is a portmanteau of coder-decoder or, less commonly, compressor-decompressor.

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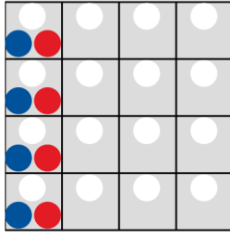
VIDEO COMPRESSION 288

Chrominance sub-sampling is nearly always applied before any compression.

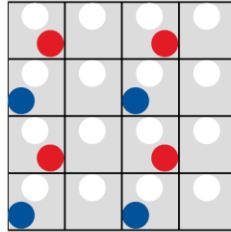
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VIDEO COMPRESSION 289



*4:1:1 chrominance
sub-sampling*



*4:2:0 chrominance
sub-sampling*

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VIDEO COMPRESSION 290

Spatial compression of individual video frames is usually based on a Discrete Cosine Transformation, like JPEG.

DV compression is purely spatial. It extends the JPEG technique by using a choice of sizes for transform blocks, and by shuffling, to even out change across a frame.

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VIDEO COMPRESSION 291

Temporal compression works by computing the difference between frames instead of storing every one in full.

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VIDEO COMPRESSION 292



Frame difference

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VIDEO COMPRESSION 293

In MPEG terminology, I-pictures are only spatially compressed. P-pictures are computed from a preceding I- or P-picture.

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VIDEO COMPRESSION 294

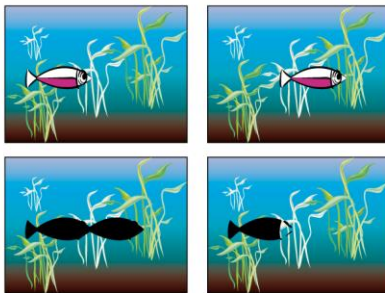
Motion compensation is the technique of incorporating a record of the relative displacement of objects in the difference frames, as a motion vector.

In existing codecs, motion compensation is applied to macroblocks, since coherent objects cannot usually be identified.

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VIDEO COMPRESSION 295



Motion compensation

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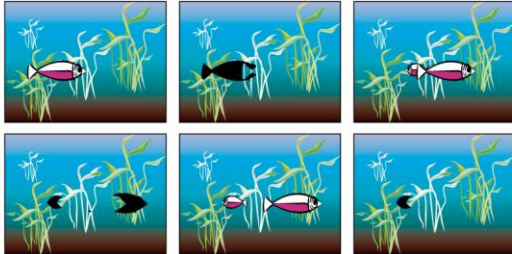
VIDEO COMPRESSION 296

B-pictures use following pictures as well as preceding ones as the basis of frame differences and motion compensation.

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VIDEO COMPRESSION 297



Bi-directional prediction

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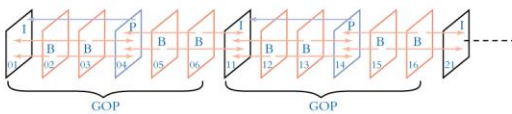
VIDEO COMPRESSION 298

A video sequence is encoded as a Group of Pictures (GOP). If B-pictures are used, a GOP may have to be reordered into display order for decoding.

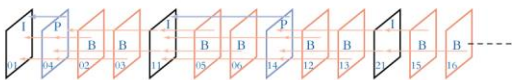
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VIDEO COMPRESSION 299



An MPEG sequence in display order



An MPEG sequence in bitstream order

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VIDEO COMPRESSION 300

MPEG-4 Part 2 uses global motion compensation and sub-pixel motion compensation to improve on the quality of MPEG-1 and MPEG-2.

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VIDEO COMPRESSION 301

H.264/AVC adds several extra techniques, including variable-sized transform blocks and macroblocks, and a de-blocking filter, to make further improvements.

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VIDEO COMPRESSION 302

Windows Media 9 (standardized as VC-1) incorporates similar improvements.

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VIDEO COMPRESSION 303

On2 VP6 and Ogg Theora are less powerful, but widely or freely available.

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VIDEO COMPRESSION 304

All modern codecs produce excellent quality at 2 Mbps and higher.

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VIDEO COMPRESSION 305



Compressed video at high quality

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EDITING AND POST-PRODUCTION 306

Editing and Post-Production

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EDITING AND POST-PRODUCTION 307

Video editing is the process of constructing a complete movie from a set of video clips or scenes, combining them with sound where required.

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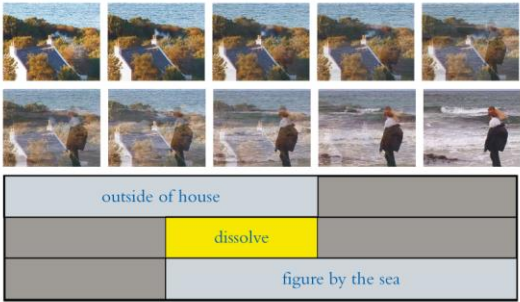
VIDEO

EDITING AND POST-PRODUCTION 308

Post-production is concerned with making changes or compositing the material, using operations that are similar to bitmapped image manipulations.

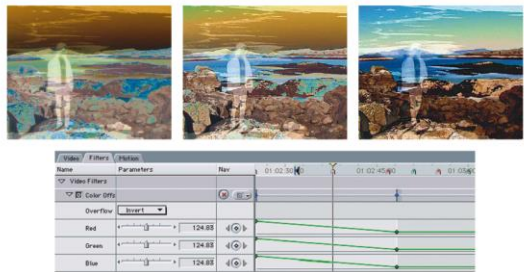
SMPTE timecode is used to identify frames by their time coordinates.

During editing, clips are imported, trimmed and assembled on a timeline. Transitions, such as dissolves, may be added between overlapping clips.



Overlapping clips for a transition

In post-production, the values of effects' parameters may vary over time.



A colour offset filter applied to video over time

Chroma keying, luma keying and mattes are used when combining tracks.

Delivery

Video may be delivered over a network as a downloaded file, it may be streamed or it may be delivered by progressive download.

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VIDEO

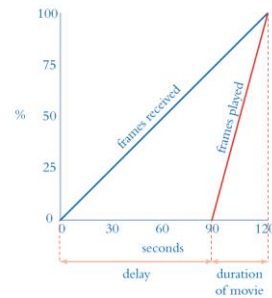
DELIVERY 317

Progressive download means that the movie starts playing when the time taken to download the remaining frames is less than the time it will take to play the whole movie.

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DELIVERY 318



Progressive download

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DELIVERY 319

When video is streamed, each frame is played as it arrives.

Streaming allows live video and does not require a file to be saved on the user's disk, but it does require sufficient bandwidth to deliver frames fast enough to be played.

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DELIVERY 320

A multimedia architecture provides an API for capture, playback and compression; a container format; a streaming server; and software tools, such as a player.

An Application Programming Interface (API) is a set of functions, procedures, methods or classes used by computer programs to request services from the operating system, software libraries or any other service providers running on the computer. A computer programmer uses the API to make application programs.

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DELIVERY 321

QuickTime and DirectShow are the multimedia architectures included with Mac OS X and Windows, respectively. Their file formats are MOV and WMV.

QuickTime and DirectShow are extensible by way of components, which allow them to use many codecs, including H.264/AVC and WMV 9, and to read and write several additional file formats, such as MP4 and AVI.

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DELIVERY 322

Flash Video is widely used for Web video. FLV files must be played in the Flash Player with a SWF that controls the video playback (or in a third-party player).

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DELIVERY 323

Ogg is an open format, that can be used in conjunction with the Theora codec to produce movies that are not subject to any restrictions or licence fees.

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VIDEO

DELIVERY 324

Web video may need its frame size and frame rate reduced before it is compressed and exported to a suitable format (e.g. MP4 or FLV) which can be played back on most systems.